

Cryptography

CSIT — 5th Semester — 2079 — Answer Sheet
Subject Code: CSC327

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Illustrate the concept of security policy and mechanism with an example. Differentiate between block cipher and stream cipher. Explain the process of key expansion in AES.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Describe the properties of hash functions. Discuss how hash value is generated using SHA-1 algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Show that Z_{25} is a field. John publishes the ElGamal public key $(q, \alpha, Y_a) = (101, 2, 14)$. Jane desired to send the secret message "CSIT" to John. Using the equivalence $A:0, B:1, \dots, Z:25$, encrypt the message using John's public key. Use a random integer $k = 4$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Differentiate between Trojan horse and virus. Describe any two types of intruders.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. The message "IMOGUN" was encrypted with a Playfair cipher using keyword "GALOIS". Decrypt the message.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. How encryption is done using IDEA algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Describe the services provided by Pretty Good Privacy protocol to secure email.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What is the condition for two integers, x and y , to be relatively prime? Find whether 61 is prime or not using Miller-Rabin algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Define challenge response system. Why do we need Kerberos?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. How direct digital signature is different from arbitrated digital signature? How digital signature generation and verification is done using RSA.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Which one is more secure, monoalphabetic cipher or poly alphabetic cipher? Justify. Using Rail Fence cipher, encrypt the text "LEARNING AND TEACHING ARE DIFFERENT", using 3 as rails.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Why do we need discrete logarithm over normal logarithm? Find out whether 3 is primitive root of 7 or not.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.